HOSPITAL INSIGHTS

Hospital Operations, Cost, and Patient Flow Intelligence Report

1. Introduction and Background

The healthcare organization seeks to gain a deeper understanding of its hospital operations, encounter costs, payer coverage, procedure utilization, and patient activity. This analysis examines the operational and financial realities of the hospital through encounter volumes, healthcare expenditure, payer contributions, procedure costs, and patterns in patient visits.

The primary objective is to extract actionable business intelligence that can help optimize hospital operations, improve visibility into encounter and procedure costs, strengthen financial oversight, and support more effective resource allocation. This will help the organization understand how healthcare services are utilized, where resources are spent, and which operational areas require management attention.

2. Key Business Questions

Financial Risk Distribution: What proportion of hospital encounter costs is covered by third-party payers, and what proportion is associated with patients without recorded payer coverage?

Payer Performance: How do encounter volume and cost differ between covered and non-covered patient groups?

High-Cost Driver Identification: Which medical procedures and encounter classes account for the largest share of healthcare expenditure?

Resource Allocation: Which clinical services experience the highest patient volumes and may require greater staffing, equipment, and operational resources?

Cost Drivers: Are expenditures concentrated in a small number of expensive procedures or distributed across many services?

Clinical and Operational Patterns: How do encounter volumes and procedure utilization vary over time and across patient demographics?

Patient Activity and Return Visits: What patterns can be identified in patient activity, and does the data support an assessment of repeat visits?

3. Key Business Insights

3.1 Overall Hospital Activity and Financial Exposure

The dashboard records approximately **27,891 encounters** and **47,701 procedures**. Total encounter claim cost is approximately **$101.514 million**, while total procedure base cost is approximately **$105.52 million**. It also reports approximately **$31.10 million in total payer coverage** and **$3.24 million in base encounter cost**. These figures demonstrate the scale of hospital activity and the financial resources associated with delivering care. Management should regularly monitor encounter expenditure, procedure costs, and payer coverage.

3.2 Ambulatory Care Accounts for the Largest Share of Encounters

Ambulatory services account for approximately **12,540 encounters (44.95%)**. Urgent care follows with approximately **6,300 (22.59%)**, while outpatient services account for approximately **3,670 (13.14%)**. Emergency services contribute 8.33%, wellness 6.92%, and inpatient services make up the remainder. Ambulatory and urgent care therefore require adequate staffing, equipment, and operational capacity.

3.3 Ambulatory and Urgent Care Account for a Large Share of Encounter Costs

Ambulatory care contributes approximately **$36.28 million (35.74%)** of encounter claim costs. Urgent care follows with **$23.35 million (23%)**, and outpatient services account for **$14.09 million (13.88%)**. Emergency services contribute about $10.75 million, inpatient services $8.11 million, and wellness $8.23 million. Management should investigate the major cost components of these high-volume services.

3.4 Payer Coverage Is Not Recorded for a Substantial Proportion of Encounters

Approximately **14,310 encounters (51.29%)** are classified as covered, while **13,590 (48.71%)** fall under no cover. Base encounter costs are approximately $1.69 million for covered encounters and $1.56 million for encounters without coverage. The hospital should investigate coverage gaps and strengthen patient financial counselling and payment processes.

3.5 Most Encounters Are Completed Within 24 Hours

Approximately **99.79%** of encounters fall under 24 hours, while **0.21%** exceed 24 hours. Recorded encounters are therefore predominantly short-duration visits. Management should maintain efficient patient flow and review longer encounters to determine whether they involve clinical complexity or operational delays.

3.6 Procedure Expenditure Is Concentrated in Certain Medical Services

Electric Cardioversion is the largest displayed procedure-cost contributor, accounting for approximately **$35.82 million (33.95%)** of total procedure base costs. The average base-cost visual also identifies procedures such as Administer ICU as having high average costs. High-cost procedures should be reviewed for equipment, consumables, staffing, and other resource requirements.

3.7 Procedure Activity Varies Across the Years

The procedure and encounter charts show changes in activity from 2011 to 2022. Procedure volumes peak around **2014**, with approximately 6,300 recorded encounters associated with procedures, before declining and fluctuating. Encounter volumes also vary, with notable peaks around 2014 and 2021. Management should investigate changes in patient demand, service availability, clinical capacity, and data completeness.

3.8 Procedure Costs Are Relatively Balanced Across Gender

The procedure-cost breakdown shows approximately **$55.42 million (52.52%)** associated with male patients and **$50.10 million (47.48%)** with female patients. Male patients account for a slightly larger share. Demographic analysis can monitor utilization, although clinical explanations require further investigation.

3.9 Patient Encounter Volumes Differ Across Clinical Categories

Ambulatory services account for the largest number of encounters, followed by urgent care and outpatient services. Ambulatory encounters also account for the largest share of total claim costs. Management should allocate staffing, equipment, and budgets according to both patient volume and service cost.

4. Business Risks

4.1 High Proportion of Encounters Without Recorded Payer Coverage

Approximately 48.71% of encounters fall under no cover. This may increase reliance on direct patient payments and expose the hospital to collection difficulties. Management should investigate coverage gaps and improve insurance-information verification and communication of payment responsibilities.

4.2 High Concentration of Procedure Expenditure

Electric Cardioversion accounts for approximately $35.82 million, or 33.95% of displayed procedure base costs. This concentration creates financial exposure if costs increase or reimbursement is inadequate. Management should review its cost structure and reimbursement arrangements.

4.3 Significant Encounter Costs in High-Volume Services

Ambulatory and urgent care together account for approximately 58.74% of encounter claim costs. Expenditure may pressure budgets if volumes rise without improved efficiency. Management should monitor cost per encounter, staffing, and resource utilization.

5. Business Opportunities

5.1 Strengthen Payer Coverage and Financial Management

The 48.71% of encounters classified as no cover creates an opportunity to improve insurance verification, payer registration, patient financial counselling, and payment arrangements.

5.2 Improve Cost Management for High-Expenditure Procedures

Electric Cardioversion is a major contributor to procedure expenditure. Reviewing procurement, equipment utilization, consumables, and procedure-related resources may reveal cost efficiencies.

5.3 Optimize Ambulatory and Urgent Care Operations

These services account for the largest shares of encounter volumes and costs. Better scheduling, staffing, patient flow, and resource planning could help the hospital manage demand more efficiently.

6. Business Recommendations

6.1 Strengthen Insurance Verification and Payer Coverage Processes

Improve insurance verification during registration and confirm payer information before services are provided where appropriate. Give patients without recorded coverage clear information about payment responsibilities and available arrangements.

Support: 48.71% of encounters are classified as no cover. **Benefit:** Better payer information, financial transparency, and potentially stronger collections. **Measures:** Verified payer-information rate, covered-encounter rate, and outstanding patient balances.

6.2 Conduct Detailed Cost Reviews of High-Expenditure Procedures

Review high-cost procedures, particularly Electric Cardioversion, examining equipment, consumables, staffing, procurement, and other resources.

Support: Electric Cardioversion represents approximately $35.82 million (33.95%) of displayed procedure base costs. **Benefit:** Improved cost visibility and procurement decisions. **Measures:** Procedure cost, average cost per procedure, and annual cost changes.

6.3 Optimize Ambulatory and Urgent Care Operations

Assess staffing, equipment, scheduling, and patient flow in ambulatory and urgent care, allocating resources according to demand and service requirements.

Support: Ambulatory care accounts for 44.95% of encounters and urgent care 22.59%; together they contribute about 58.74% of encounter claim costs. **Benefit:** Improved patient flow and resource utilization. **Measures:** Average encounter duration, encounters per department, and cost per encounter.

6.4 Monitor Encounter Costs Across Clinical Categories

Compare total costs, patient volumes, and average costs per encounter for each encounter class to identify services where expenditure rises disproportionately to activity.

Support: Ambulatory and urgent care account for substantial shares of encounter costs. **Benefit:** Better budget control and planning. **Measures:** Cost per encounter, total cost by encounter class, and monthly cost variation.

6.5 Investigate Changes in Encounter and Procedure Volumes

Investigate annual fluctuations in activity, including service demand, staffing capacity, equipment availability, and changes in clinical operations.

Support: Trend charts show variations, with notable peaks around 2014 and 2021. **Benefit:** Better forecasting and capacity planning. **Measures:** Annual encounter growth, procedure volume, and utilization by clinical category.

6.6 Improve Patient Flow and Monitor Extended Encounters

Maintain efficient processes for shorter encounters and review encounters exceeding 24 hours to identify clinical complexity, resource constraints, or operational delays.

Support: 99.79% of encounters fall under 24 hours and 0.21% exceed 24 hours. **Benefit:** Better patient flow and service coordination. **Measures:** Share of encounters exceeding 24 hours and average encounter duration.

6.7 Introduce Regular Demographic and Procedure Utilization Reviews

Monitor procedure utilization across demographic groups and investigate differences in procedure volumes and costs to support resource planning.

Support: Male patients account for 52.52% of procedure base costs, compared with 47.48% for female patients. **Benefit:** Better understanding of utilization patterns. **Measures:** Procedure volume and average procedure cost by gender, and procedure mix by demographic group.

7. Evidence Limitations and Areas for Further Investigation

- **Payer coverage versus actual reimbursement:** The dashboard shows payer coverage and covered/no-cover classifications but does not establish claims approved, denied, or successfully collected. Claims and payment data are needed to assess actual payer performance and revenue recovery.
- **Payer-level performance:** The visuals do not clearly compare individual insurers. Further payer-level data is needed to identify low coverage, delayed payments, or claim rejections.
- **Patient return visits:** No direct repeat-visit rate, readmission rate, or patient-level visit sequence is displayed. A revisit analysis requires reliable patient identifiers and encounter dates.
- **Diagnosis-to-procedure relationships:** The dashboard does not establish which diagnoses generate the most procedures. Linked diagnosis and procedure data is required.
- **Procedure cost interpretation:** Average base cost and total base cost are different measures and should not be treated interchangeably. The cost basis and calculation method should be validated.
- **Encounter duration measurement:** The average visit duration is shown as 0.18, but its unit is unspecified and should be confirmed before interpretation.
- **Data completeness and time trends:** The dashboard covers 2011–2022, but does not establish whether all years contain complete and comparable records. Data availability and recording practices may influence trends.
- **Cost versus revenue:** The dashboard primarily presents claim and base costs; it does not provide enough information to establish hospital profitability or margins for individual procedures.

8. Conclusion

The analysis provides valuable insight into hospital activity, encounter expenditure, payer coverage, and procedure utilization. The dashboard records approximately **27,891 encounters and 47,701 procedures**, with total encounter claim costs of approximately **$101.514 million** and procedure base costs of approximately **$105.52 million**.

Ambulatory care is the hospital's largest encounter category, accounting for 44.95% of recorded encounters. Ambulatory and urgent care together contribute a substantial proportion of encounter costs. The payer analysis shows that 48.71% of encounters fall under the no-cover category, while Electric Cardioversion is a major contributor to procedure expenditure.

Management should prioritize payer verification, monitoring high-cost procedures, optimizing high-volume clinical services, and strengthening patient-flow and resource-planning processes. Further investigation into payer-level reimbursement, patient return visits, diagnosis-to-procedure relationships, and actual payment recovery would strengthen the analysis and support a more comprehensive understanding of hospital performance.